

S.A. ENGINEERING COLLEGE

Autonomous Institution | Affiliated to Anna University | NBA & NAAC 'A' Grade

Dept of Electronics & Communication Engg & Dept of Electronics Engg [VLSI Design & Technology]

KERNEL PRIME '26 - SOFTWARE TRACK

National Level 24-Hour Hackathon | October 8-9, 2026

OFFICIAL PROBLEM STATEMENT: PS-04

Smart Grid Renewable Energy Forecasting & Microgrid Balancing

Domain: CleanTech / Time-Series ML / Industrial IoT

1. PROBLEM OVERVIEW & CHALLENGE STATEMENT

Construct a predictive machine learning platform for sub-second solar/wind generation forecasting, dynamic battery storage dispatch, and virtual power plant dispatch under sudden load surges.

2. MANDATORY ARCHITECTURAL REQUIREMENTS

[REQ-1] Multi-horizon transformer for ultra-short-term (15-min) and day-ahead renewable power generation

[REQ-2] Mixed-integer linear programming (MILP) battery charge/discharge optimization engine.

[REQ-3] Automated peak shaving and frequency regulation command dispatch.

[REQ-4] Industrial Modbus / MQTT SCADA telemetry gateway integration.

[REQ-5] High-resolution anomaly detection for grid voltage sag, swell, and equipment degradation.

3. HACKATHON EVALUATION MATRIX

* System Architecture & Concurrency Resilience: 25%

* Technical Innovation & Production Viability: 25%

* Live Demo & Real-World Latency Benchmarks: 30%

* Security Hardening, UI Polish & Documentation: 20%

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